MAINCRAFTS TECHNOLOGY

Artificial Intelligence & Machine Learning — Task 4

Technical Report: Classification Models, Evaluation Metrics & Handling Imbalanced Data

1. Classification Metrics & Confusion Matrix Interpretation

When moving from regression to classification, evaluating models solely on simple accuracy can be highly misleading, especially in the presence of class imbalance. A breakdown of performance metrics using the Breast Cancer Dataset (0: Malignant, 1: Benign) explains why secondary metrics are necessary.

1.1 The Confusion Matrix Breakdown

The confusion matrix is a tabular layout that categorizes the model's predictions against the actual ground-truth labels:

- **True Positives (TP):** The model correctly predicts the positive class (e.g., correctly identifying a patient with a Malignant tumor).
- **True Negatives (TN):** The model correctly predicts the negative class (e.g., correctly identifying a Benign sample).
- **False Positives (FP):** The model incorrectly flags a negative sample as positive (Type I error: telling a patient with a Benign tumor that it is Malignant).
- **False Negatives (FN):** The model incorrectly flags a positive sample as negative (Type II error: telling a patient with a Malignant tumor that it is Benign).

1.2 Why Accuracy Alone is Insufficient

Accuracy is calculated as:

$$\text{Accuracy} = \frac{TN+TP}{TN+TP+FP+FN}$$

If a dataset contains 95% Benign samples and only 5% Malignant samples, a naive classifier that predicts "Benign" for every single patient will achieve a 95% accuracy score. Despite the high score, the model is completely useless because it fails to detect a single malignant tumor, showing why accuracy cannot be trusted on its own in critical real-world deployments.

2. Precision, Recall, and F1-Score Deep Dive

To overcome the limitations of accuracy, three business-critical metrics must be analyzed:

2.1 Medical Diagnosis Metric Priority

In healthcare domains, such as breast cancer screening, **Recall** is prioritized over Precision.

- **The Impact of Low Recall:** If a model's recall is low, it means there is a high count of False Negatives. In medical terms, malignant tumors are missed, sending sick patients home without treatment, which can lead to severe real-world consequences.
- Conversely, a False Positive (low precision) merely results in a follow-up diagnostic test, making it a much safer clinical error than a missed diagnosis.

2.2 The Role of F1-Score in Imbalanced Datasets

The **F1-Score** is the harmonic mean of Precision and Recall:

$$\text{F1-Score} = 2 * \frac{Precision * Recall}{Precision + Recall}$$

Unlike the arithmetic mean, the harmonic mean penalizes extreme imbalances between the two metrics. If either Precision or Recall drops near zero, the F1-score collapses, making it an excellent single metric for evaluating imbalanced production classification systems.

3. Class Imbalance Strategies & Model Comparison

3.1 Handling Imbalanced Data via Class Weights

To prevent the classifier from ignoring the minority class, the `class_weight="balanced"` parameter can be added to the Logistic Regression model. This technique adjusts the loss function weights inversely proportional to class frequencies in the input data:

$$\text{Wj} = \frac{N}{K * Nj}$$

Where N is total samples, k is number of classes, and n_j is the number of samples in class j. This artificially penalizes mistakes made on the minority class, significantly reducing False Negatives and driving up the model's Recall score.

3.2 Logistic Regression vs. Decision Tree Classifier

The table below contrasts the baseline model behavior observed across the execution of Step 5, Step 9, and Step 10.

Model Variant	Structural Properties	Overfitting Tendency	ROC-AUC Performance
Baseline Logistic Regression	High Interpretability, linear decision boundaries.	Low. Relies on smooth regularization penalties.	High baseline performance.
Balanced Logistic Regression	Identical structure but penalized loss optimizations.	Low. Balanced across minority classification boundaries.	**Optimized.** Maximizes true positive rate tracking.
Decision Tree Classifier	High Interpretability via hierarchical logic nodes.	**High.** Prone to memorizing arbitrary training sample splits.	Moderate; dependent on split depth metrics.

4. Final Model Decision Justification

The **Balanced Logistic Regression** model is selected for final deployment based on the following engineering trade-offs:

1. **Imbalance Mitigation:** By adjusting class weights, the model actively prevents minority class suppression, protecting the system against misleading accuracy scores.
2. **Generalization and Stability:** Unlike the `DecisionTreeClassifier`, which builds highly complex step-like boundaries prone to high variance, Logistic Regression scales smoothly using maximum likelihood estimation, minimizing the risk of overfitting on validation sets.
3. **Discrimination Power:** The ROC Curve and Area Under the Curve (AUC) verify that the probability outputs generated by the balanced linear model maintain cleaner class separation across various thresholds compared to unconstrained tree alternatives.